

BCCE 2026

What chemistry faculty believe about doctoral education

A phenomenographic and thematic characterization

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Doctoral education produces the chemists
who staff the discipline.

~2,900
chemistry PhDs each year

~196
U.S. doctoral programs

decades
one largely unchanged structure

Yet the same record documents its strains — competing responsibilities, no standard assessment of outcomes, uneven implementation, and inconsistent mentorship.

Beliefs about advising a PhD have never been characterized.

Classroom teaching

Beliefs are well characterized —
and predict what happens in class

Research advising

A different practice — and
never characterized

Donkor et al. (2024) set faculty beliefs aside — a proper investigation “*warranted its own study.*” **This is that study.**

RESEARCH QUESTION

What do chemistry faculty believe
about doctoral education?

what it's for

how students learn

who develops which skill

the structure they work in

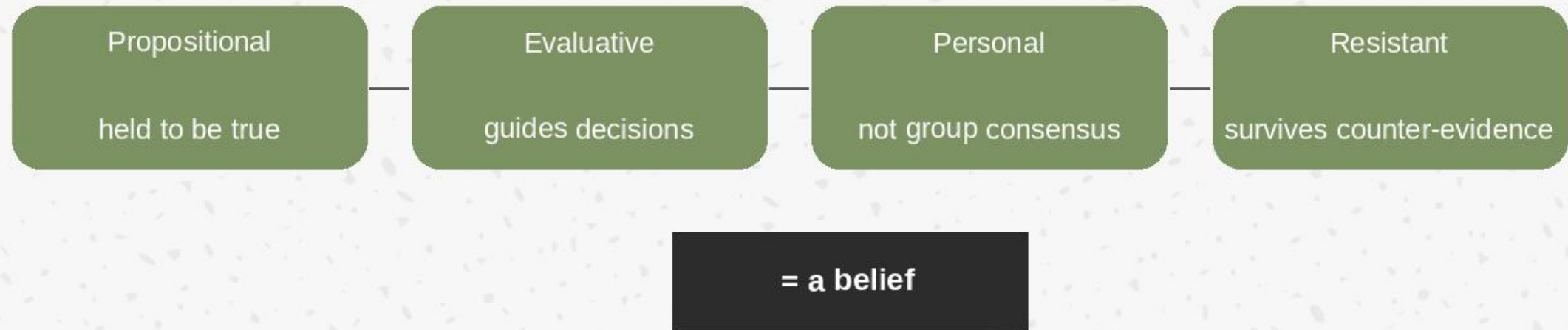
Whether a reform reaches practice depends on what teachers believe.



All four shape whether a reform reaches practice — the model's ultimate test.

Beliefs are self-perpetuating — they can absorb a reform without yielding to it.

A belief is a proposition held true
that guides how a person acts.



An interpretive design built to capture the range of faculty belief.



No new codes appeared after interview 7 — the result held from 10 participants to all 20.

Recruited for maximum variation
across profile and subfield.

N = 20
participants

13 / 7
men / women

13 / 7
White / non-White

15
full professors

14
at R1 institutions

18
U.S.-trained

Subdisciplines: Inorganic (5) · Chemistry education (2) · Materials (2) · Physical (2) · Biophysical (2) · Organic (2) · + five singletons

Analytical generalization and transferability, not population estimates; the sample skews senior and male.

Faculty agree on the propositions —
and divide on where the boundaries fall.

AGREE
on the four shared propositions

DIVIDE
on where the boundaries fall

“What differs is not which beliefs faculty hold, but where they draw the lines those beliefs require.”

The doctorate develops a self-directed scientist
in service of the student's own goals.

When they start bringing ideas forward without necessarily being prompted.

— P2 · on what independence looks like

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WHERE THEY DIVIDE

answering existing questions

asking new questions

The realistic goal versus the higher bar for the degree.

Students learn by doing scaffolded research —
and learning shows in performance, not grades.

I do. We do. You do — spread out over X number of years.

— P5 · on how research is learned

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LO

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WHERE THEY DIVIDE

unsupported struggle helps

struggle should be scaffolded

Same formative deprivation, opposite lessons — P13 vs. P7.

Development is a shared responsibility in a two-way relationship — and who develops which skill varies.

I go from the leader to the follower, and they start as the follower and end up as the leader.

— P6 · on how responsibility transfers



WHERE THEY DIVIDE

faculty's to teach

student's to build

The same skill — creativity — is one faculty's to teach (P5, P17), another's for the student (P9, P11).

Structure shapes advising — and the gaps it leaves
are the program's to fix, not the student's.

Those professional skills often get kind of forgotten.

— P2 · on the program's gap

PS

GA

WHERE THEY DIVIDE

the evolved system works

curricula & funding need change

How much of the structure deserves changing — P14 vs. P18.

Within each shared theme, faculty differ along the axes below.
Each axis names the two poles of variation; nothing is plotted on it.



THE FULL OUTCOME SPACE

Four shared themes.
Fifteen codes.
An axis of division
in every one.

The agreement is real — and so is the disagreement one level down. Every faculty member lands somewhere on each axis.

A belief baseline for changing doctoral education.

Content for the model

Gives the Teacher-Centered Systemic Reform model the doctoral-education content it lacked — the shared ground and the fault lines a reform must engage.

Nests with prior work

The goals record and the challenge record now sit beside what faculty believe. Three answers to one enterprise — they nest, not compete.

Conviction vs. practice

Thirteen of twenty hold stable convictions but refine implementation with experience — so supports aimed at implementation may fit faculty best.

Two levels, two lessons.

SHARED PROPOSITIONS

Common ground a reform can take as given — arguing from where faculty stand.

BOUNDARY DISAGREEMENTS

The harder, more useful target — where aligned colleagues quietly diverge.

Thank you.

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